

CSE AUT 23 FINAL QUISTION SOLUTION



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(ans 1 a)

----->

n = 3

3, n = 3

4, n = 4

6, n = 6

9, n = 9

(ans c)

----->

```
#include<stdio.h>
```

```
int main(){
```

```
    int i,j,n,m,x;
```

```
    scanf("%d%d",&n,&x);
```

```
    double y=(n/x);
```

```
    for(i=1;i<=y;i++){
```

```
        if(i%2!=0){
```

```
            for(j=1;j<=x;j++){
```

```
                printf("%d ",j);
```

```
            }
```

```

    }
    else{
        for(j=x;j>=1;j--){
            printf("%d ",j);
        }

    }
}
return 0;
}

```

(c a n s o r)

```

#include<stdio.h>

int main(){
    int i,j,n,x=0,k=2,count=0;

    int y;

    scanf("%d",&n);

    for(i=1;i<=n;i++){
        count=0;
        scanf("%d", &y);
        k=2;
        while(k>0){
            k=y/2;
            x=y%2;

```

```

if(x==1){
    count++;
}

y=k;
}
y=0;
if(count%2==0){
    printf("evil number\n");
}else{
    printf("odious number\n");
}
}
return 0;
}

```

(2 a answer)

_____>

Local Variable: The variable whose scope lies inside a function or a block in which they are declared.

Global Variable: The variable that exists outside of all functions. It is the variable that is visible from all other scopes.

```

#include<stdio.h>

\\global variable
int x=50;

int main(){
    \\local variable
    int x=10;

    return 0;
}

```

```
}
```

(answer 2c)

_____>

answer is === (120).

(answer 2c)

_____>

```
#include<stdio.h>
```

```
int printRang(int x,int y){
```

```
int i,sum=0;
```

```
for(i=x;i<=y;i++){
```

```
    printf("%d ",i);
```

```
    sum+=i;
```

```
}
```

```
printf("\n");
```

```
return sum;
```

```
}
```

```
int main(){
```

```
    int a,b;
```

alternative)--->line 1, 2

```
//scanf("%d%d",&a,&b);
```

```
//printf("%d",printRang(a,b));
```

```

1. int x=printRang(11,15);
2. printf("%d",x);
   return 0;
}

```

(2 c or answer)

----->

```

#include<stdio.h>

int mid(int n, int arr[]){
    int i,j,temp=0,x=((n-1)/2),y=(n/2);
    double y1;
    for(i=0;i<n;i++){
        for(j=i+1;j<n;j++){
            if(arr[i]>arr[j]){
                temp=arr[i];
                arr[i]=arr[j];
                arr[j]=temp;
            }
        }
    }
    if(n%2 != 0){
        printf("the median is %d\n",arr[x]);
        int k=arr[x];
        return k;
    }else{
        y1=((arr[y]+arr[y-1])/2.0);
        printf("the median is%.2lf\n",y1);
    }
}

```

```

        return y1;
    }
}

int main(){

    int i,a[100],x;
    int *p;
    scanf("%d",&x);
    for(i=0;i<x;i++){
        scanf("%d",&a[i]);
    }

    mid(x,a);

    return 0;
}

```

(answer 3a)

_____>

```

#include<stdio.h>

int main(){

```

```

int i ,arr[10]={0,0,0,1,4,7,10,0,0,0};

for(i=9;i>=0;i--){
printf("%d ",arr[i]);
}

return 0;
}

```

(answer 3 b)

----->

```

#include<stdio.h>

int main(){
    int n,i,f;
    scanf("%d",&n);
    int arr[n];
    for(i=0;i<n;i++){
        scanf("%d",&arr[i]);
    }
    for(i=0;i<n-2;i++){
        if(arr[i]==1 && arr[i+1]==1 && arr[i+2]==1){
            f=1;
            break;
        }
    }
    if(f==1){
        printf("yes\n");
    }else{

```

```
    printf("no\n");  
}  
return 0;  
}
```

(answer 3 c)

_____>

```
#include<stdio.h>  
#include<string.h>  
int main(){  
  
    char arr[1000];  
    int i,j,x,f;  
    gets(arr);  
    x=strlen(arr);  
  
    for(i=0,j=x-1;i<j;i++,j--){  
        if(arr[i]==arr[j]){  
            f=1;  
        }else{  
            f=0;  
        }  
    }  
    if(f==1){  
        printf("yes palindrom\n");  
    }  
}
```



```
}else{  
    printf("not palindrom\n");  
}  
  
return 0;  
}
```

(answer 3c or)

_____>

```
#include<stdio.h>  
  
int main(){  
    int i,j,x,n,arr[10000],f=0,k;  
    scanf("%d%d",&n,&x);  
    for(i=0;i<n;i++){  
        scanf("%d",&arr[i]);  
    }  
  
    for(i=0;i<x-1;i++){  
        if(arr[i]<arr[i+1]){  
            f=1;  
        }else{  
            f=0;  
        }  
    }  
  
    printf("%d\n",f);  
    if(f==1){
```

```
for(i=x;i<n-1;i++){  
    if(arr[i]>arr[i+1]){  
        f=1;  
  
    }else{  
        f=0;  
    }  
}  
}
```

```
if(f==1){  
    printf("yes\n");  
}else{  
    printf("no\n");  
}
```

```
return 0;  
}
```

(answer 4 a)

----->

```
#include<stdio.h>  
  
int main(){  
  
    int i,j,row,col,s,cnt=0,sum=0;
```

```
int f=0;
```

```
int arr[100][100];
```

```
scanf("%d%d%d",&row,&col,&s);
```

```
for(i=0;i<row;i++){
```

```
    for(j=0;j<col;j++){
```

```
        scanf("%d",&arr[i][j]);
```

```
    }
```

```
}
```

```
for(i=0;i<row;i++){
```

```
    f=0;
```

```
    for(j=0;j<col;j++){
```

```
        if(arr[i][j]==s){
```

```
            f=1;
```

```
        }
```

```
    }
```

```
    if(f==1)cnt++;
```

```
}
```

```
printf("%d",cnt);
```

```
return 0;
```

```
}
```

(answer 4 b)

----->

1) not valid 2)vald 3)not valid 4) valid

(4c answer)

----->

```
#include<stdio.h>
#include<string.h>
int main(){
int i,j,x,c=0,n=0,ch=0;
char arr[20],cmp[20]="12345678";
scanf("%s",arr);
x=strlen(arr);
if(x>=8){
if(strcmp(cmp,arr)!=0){
for(i=0;arr[i]!=0;i++){
if(arr[i]>='A' && arr[i]<='Z'){
c++;
}else if(arr[i]>='0' && arr[i]<='9'){
n++;
}else {
ch++;
}
}
}
if(c>0 && n>0 && ch>0){
printf("valid password\n");
}
```

```

    }else{
        printf("not valid\n");
    }
}
else{
    printf("not valid\n");
}

return 0;
}

```

(4 c or)

----->

```

#include<stdio.h>
#include<string.h>
int main(){
    char arr[100],ayy[100],ajj[300];
    scanf("%s %s",arr,ayy);
    int i,count=0,count1=0,f=0;
    for(i=0;arr[i]!=0;i++){
        count++;
    }

    for(i=0;ayy[i]!=0;i++){
        count1++;
    }

    int j=0;

```

```

for(i=count-1;arr[i]>=0;i--){
    ajj[j]=arr[i];
    j++;
}

if(count==count1){
for(i=0;ajj[i]!=0;i++){
    if(ajj[i]==ayy[i]){
        f=1;
    }else{
        f=0;
    }
}
}else{
    printf("not same\n");
}

if(f==1){
    printf("accurate\n");
}else{
    printf("not same\n");
}

return 0;
}

```

(5 a ansewr)

----->

pass by value-->

when we pass the value of the parameter during the calling of the function, it copies them to the function's actual local argument.

```
#include <stdio.h>
```

```
void swapx(int x, int y);
```

```
int main()
```

```
{
```

```
    int a = 10, b = 20;
```

```
    swapx(a, b);
```

```
    return 0;
```

```
}
```

```
void swapx(int x, int y)
```

```
{
```

```
    int t;
```

```
    t = x;
```

```
    x = y;
```

```
    y = t;
```

```
    printf("Inside Function:\nx = %d y = %d\n", x, y);
```

```
}
```

pass by reference---->

In the case of Call by Reference, when we pass the parameter's location reference/address,

it copies and assigns them to the function's local argument.

```
void swapx(int*, int*);

int main()
{
    int a = 10, b = 20;
    swapx(&a, &b);
    return 0;
}

void swapx(int* x, int* y) {
    int t;

    t = *x;
    *x = *y;
    *y = t;

    printf("Inside the Function:\nx = %d y = %d\n", *x, *y);
}
```

(answer 5 b)

----->

```
#include<stdio.h>
#include<string.h>
struct Bookinfo{
    char title[99],authername[49];
    double price;
};
```



```

typedef struct Bookinfo std;

std arr[1000];

int main(){
    int i,j,n;
    scanf("%d",&n);
    for(i=0;i<n;i++){
        scanf("%s %s %lf",arr[i].title,arr[i].authorname,&arr[i].price);
    }
    for(i=0;i<n;i++){
        printf("%s\n%s\n%.2lf\n",arr[i].title,arr[i].authorname,arr[i].price);
    }
    return 0;
}

```

(answer 5c)

----->

```

#include <stdio.h>
#include <string.h>
int currentscore = 0;
int main()
{
    FILE *fr;
    fr = fopen("highscore.txt", "r");
    fscanf(fr, "%d", &currentscore);
    fclose(fr);
}

```

```
int nws = 0;

scanf("%d", &nws);

if (nws > currentscore)
{
    fr = fopen("highscore.txt", "w");
    fprintf(fr, "%d", nws);
    fclose(fr);
}
return 0;
}
```

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